

Distributional Reinforcement Learning with Function Approximation

Abstract

These notes give an overview of some theoretical foundations behind approximation based distributional reinforcement learning and newer methods using differentiable metrics on probability measures. We begin with linear value-function approximation in the standard scalar case, then we replace scalar values by return distributions and introduce the two classic representations, quantile and categorical, of probability distributions. Then, we go over how these representations can be used in function approximation using deep neural networks. Finally, we cover the recent research on distributional reinforcement learning using differentiable metrics on probability measures such as the MMD distances or Sinkhorn divergence.

Preliminaries

The first part of these notes follow the treatment of distributional reinforcement learning in [BDR23]. We first recall some of the preliminary definitions and results.

Definition 1: Finite Markov decision process and policy

A finite Markov decision process is given by

$$(\mathcal{X}, \mathcal{A}, \xi_0, P_X, P_R),$$

where $\mathcal{X}$ is a finite state space, $\mathcal{A}$ is a finite action space, ξ_0 is an initial-state distribution, $P_X(\cdot \mid x, a)$ is the next-state kernel, and $P_R(\cdot \mid x, a)$ is the reward distribution. A policy is a map

$$\pi : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{A}), \quad x \mapsto \pi(\cdot \mid x),$$

and the discount factor satisfies $\gamma \in [0, 1)$. The markov chain induced by P_X and π gives rises to the trajectory measure P_π .

Given a policy π , the trajectory process is

$$\begin{aligned} X_0 &\sim \xi_0, & A_t &\sim \pi(\cdot \mid X_t), \\ R_t &\sim P_R(\cdot \mid X_t, A_t), & X_{t+1} &\sim P_X(\cdot \mid X_t, A_t), & t \geq 0. \end{aligned}$$

Starting from $X_0 = x$ and then following π , define the discounted return

$$G^\pi(x) = \sum_{t=0}^{\infty} \gamma^t R_t.$$

Definition 2: Value function and return distribution

The state-value function is the expected return

$$V^\pi(x) = \mathbb{E}_\pi[G^\pi(x)].$$

The return distribution at state x is

$$\eta_\pi(x) = \text{Law}(G^\pi(x)).$$

Thus the scalar value function is the first moment of the return distribution:

$$V^\pi(x) = \int_{\mathbb{R}} z \eta_\pi(x)(dz).$$

Definition 3: Bellman operator for scalar values

For a function $V : \mathcal{X} \rightarrow \mathbb{R}$, the Bellman operator for policy π is

$$(\mathcal{T}^\pi V)(x) = \mathbb{E}_{A \sim \pi(\cdot | x)} [\mathbb{E} [R + \gamma V(X') \mid X = x, A]].$$

Equivalently, the outer expectation samples the action from the policy and the inner expectation samples reward and next state from the MDP. The Bellman equation is

$$V^\pi = \mathcal{T}^\pi V^\pi.$$

The scalar Bellman equation comes from a stronger identity at the level of random variables. With

$$A \sim \pi(\cdot \mid x), \quad R \sim P_R(\cdot \mid x, A), \quad X' \sim P_X(\cdot \mid x, A),$$

the return satisfies

$$G^\pi(x) \stackrel{D}{=} R + \gamma G^\pi(X') \quad \text{given } X = x.$$

Taking laws gives

$$\eta_\pi(x) = \text{Law}(R + \gamma G^\pi(X') \mid X = x).$$

Definition 4: Distributional Bellman operator

For $r, z \in \mathbb{R}$, define the bootstrap map

$$b_{r,\gamma}(z) = r + \gamma z.$$

If $\nu \in \mathcal{P}(\mathbb{R})$ and $Z \sim \nu$, then the pushforward $(b_{r,\gamma})_{\#}\nu$ is the law of $r + \gamma Z$:

$$(b_{r,\gamma})_{\#}\nu = \text{Law}(r + \gamma Z).$$

For a return-distribution function $\eta : \mathcal{X} \rightarrow \mathcal{P}(\mathbb{R})$, the distributional Bellman operator is

$$(\mathcal{T}^{\pi}\eta)(x) = \mathbb{E}_{A \sim \pi(\cdot|x)} [\mathbb{E} [(b_{R,\gamma})_{\#}\eta(X') \mid X = x, A]].$$

The return-distribution function η_{π} is its fixed point:

$$\eta_{\pi} = \mathcal{T}^{\pi}\eta_{\pi}.$$

Theorem 1: Wasserstein contraction of the distributional Bellman operator

For $p \in [1, \infty]$, let

$$\bar{w}_p(\eta, \nu) = \sup_{x \in \mathcal{X}} w_p(\eta(x), \nu(x)).$$

Then

$$\bar{w}_p(\mathcal{T}^{\pi}\eta, \mathcal{T}^{\pi}\nu) \leq \gamma \bar{w}_p(\eta, \nu).$$

Proof. We give the coupling proof from [BDR23, Proposition 4.15] for $p < \infty$. For each state y , take an optimal coupling $(Z(y), Z'(y))$ of $\eta(y)$ and $\nu(y)$. Couple the two Bellman updates by using the same sampled action, reward, and next state in both. Conditional on $X = x$, the resulting pair

$$\tilde{Z} = R + \gamma Z(X'), \quad \tilde{Z}' = R + \gamma Z'(X')$$

is a valid coupling of $(\mathcal{T}^{\pi}\eta)(x)$ and $(\mathcal{T}^{\pi}\nu)(x)$. Therefore

$$\begin{aligned} w_p^p((\mathcal{T}^{\pi}\eta)(x), (\mathcal{T}^{\pi}\nu)(x)) &\leq \mathbb{E}[|\tilde{Z} - \tilde{Z}'|^p \mid X = x] \\ &= \gamma^p \mathbb{E}[|Z(X') - Z'(X')|^p \mid X = x] \\ &\leq \gamma^p \sup_{y \in \mathcal{X}} w_p^p(\eta(y), \nu(y)). \end{aligned}$$

Taking the p th root and then the supremum over x proves the claim. The $p = \infty$ case uses the same coupling and the essential supremum. $\square$

Definition 5: Finite representations of return distributions

For a probability measure $\eta \in \mathcal{P}(\mathbb{R})$, let

$$F_\eta(t) = \eta((-\infty, t]), \quad F_\eta^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\eta(z) \geq \tau\}.$$

Two standard finite representations are:

$$\text{categorical:} \quad \eta(x) = \sum_{i=1}^m p_i(x) \delta_{\theta_i}, \quad p_i(x) \geq 0, \quad \sum_{i=1}^m p_i(x) = 1,$$

and

$$\text{quantile:} \quad \eta(x) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)}, \quad \tau_i = \frac{2i-1}{2m}, \quad \theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i).$$

If $\mathcal{X}$ is large, storing a scalar value for every state is already expensive; storing a full probability distribution for every state is even more so. The rest of the notes is about approximation: first for scalar values, and then for return distributions.

1 Linear Function Approximation

We first discuss linear approximation for the state-value function

$$V^\pi : \mathcal{X} \rightarrow \mathbb{R}.$$

This section follows parts of Chapter 9 of [BDR23].

1.1 Function Approximation

In linear function approximation, we represent the value function as a linear combination of features. Similar to neural networks, the features here are allowed to be nonlinear.

Definition 6: State representation and linear value approximation

Let $n \in \mathbb{N}$. A state representation is a mapping

$$\phi : \mathcal{X} \rightarrow \mathbb{R}^n, \quad \phi(x) = \begin{pmatrix} \phi_1(x) \\ \vdots \\ \phi_n(x) \end{pmatrix}.$$

For $w \in \mathbb{R}^n$, the linear value function approximation is

$$V_w(x) = \phi(x)^\top w,$$

where each ϕ_i is a basis function and $\phi_i(x)$ is a feature.

The model is linear in its weights and

$$V_{\alpha w_1 + \beta w_2} = \alpha V_{w_1} + \beta V_{w_2}, \quad \nabla_w V_w(x) = \phi(x).$$

For finite $\mathcal{X}$, collect the row vectors $\phi(x)^\top$ in a feature matrix $\Phi \in \mathbb{R}^{|\mathcal{X}| \times n}$. Then

$$V_w = \Phi w, \quad \mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}.$$

The same can be done for action value function using $\phi : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}^n$ and $Q_w(x, a) = \phi(x, a)^\top w$. Since we want to focus on what the distributional part changes about the approximation later, we will mostly discuss the state value case.

1.2 Optimal Linear Value Function Approximation

As states above, the equation

$$V_w = \Phi w$$

defines a linear system, which won't have a solution in general. Instead, a standard way to find the 'best' solution is to minimize the squared error. This can also be done here, however in [BDR23] it is mentioned that in RL we are often not equally interested in all states, hence we introduce a weighted least squares loss.

So, let $\xi \in \mathcal{P}(\mathcal{X})$ be the relative importance of the states and define

$$\|V - V'\|_{\xi, 2} = \left(\sum_{x \in \mathcal{X}} \xi(x) (V(x) - V'(x))^2 \right)^{1/2}.$$

we are thus interested in the best approximation to V^π which solves

$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi, 2}.$$

and this is given by a weighted normal equation.

Theorem 2: Weighted normal equation

Let $\Xi = \text{diag}(\xi(x) : x \in \mathcal{X})$. If ξ has full support and the columns of Φ are linearly independent, the unique solution is

$$w^* = (\Phi^\top \Xi \Phi)^{-1} \Phi^\top \Xi V^\pi.$$

Thus the state weighting ξ is part of the approximation problem: changing how often states are sampled generally changes the best attainable weights.

The problem is, we can't iterate the Bellman operator since after applying it, we might not have a linear representation anymore. This motivates the following definition of a projected Bellman operator.

1.3 The Projected Bellman Operator

Definition 7: Projected Bellman operator

Define the weighted orthogonal projection $\Pi_{\phi,\xi}$ by

$$(\Pi_{\phi,\xi}V)(x) = \phi(x)^\top w^*, \quad w^* \in \arg \min_{w \in \mathbb{R}^n} \|V - V_w\|_{\xi,2}.$$

Approximate dynamic programming iterates

$$V_{k+1} = \Pi_{\phi,\xi} \mathcal{T}^\pi V_k.$$

Luckily, one can still show that this projected Bellman operator is a contraction and that the fixed point of the projected Bellman operator is close to the best approximation of the true value function. Hence, by iteration the Bellman equation like this, we will converge to something that is close to what we are trying to approximate!

Theorem 3: Projected Bellman contraction and error bound

Let ξ_π be a fully supported steady-state distribution under π . Then Π_{ϕ,ξ_π} is a nonexpansion in the ξ_π -weighted L_2 norm. Consequently,

$$\|\Pi_{\phi,\xi_\pi} \mathcal{T}^\pi V - \Pi_{\phi,\xi_\pi} \mathcal{T}^\pi V'\|_{\xi_\pi,2} \leq \gamma \|V - V'\|_{\xi_\pi,2}.$$

The projected operator has a unique fixed point

$$\hat{V}^\pi = \Pi_{\phi,\xi_\pi} \mathcal{T}^\pi \hat{V}^\pi,$$

Additionally, they show that it holds

$$\|\hat{V}^\pi - V^\pi\|_{\xi_\pi,2} \leq \frac{1}{\sqrt{1-\gamma^2}} \|\Pi_{\phi,\xi_\pi} V^\pi - V^\pi\|_{\xi_\pi,2}.$$

which means that the fixed point of the projected Bellman operator is close to the best approximation.

Proof. The proofs follows the argument from [BDR23, Theorem 9.8]. The transition operator P^π is nonexpansive and due to the orthogonality of the projection we have for every U ,

$$U = \Pi_{\phi,\xi_\pi} U + (I - \Pi_{\phi,\xi_\pi})U$$

is an orthogonal decomposition in the ξ_π -weighted inner product. Hence, by Pythagoras,

$$\|U\|_{\xi_\pi,2}^2 = \|\Pi_{\phi,\xi_\pi} U\|_{\xi_\pi,2}^2 + \|(I - \Pi_{\phi,\xi_\pi})U\|_{\xi_\pi,2}^2,$$

so $\|\Pi_{\phi,\xi_\pi} U\|_{\xi_\pi,2} \leq \|U\|_{\xi_\pi,2}$. Thus the composition with $\mathcal{T}^\pi$ is a γ -contraction. Banach's fixed-point theorem gives a unique fixed point and convergence of iterating the projected Bellman operator.

For the error estimate, we use orthogonality of the projection and $V^\pi = \mathcal{T}^\pi V^\pi$ which yields

$$\begin{aligned}\|\hat{V}^\pi - V^\pi\|_{\xi_\pi, 2}^2 &= \|\hat{V}^\pi - \Pi_{\phi, \xi_\pi} V^\pi\|_{\xi_\pi, 2}^2 + \|\Pi_{\phi, \xi_\pi} V^\pi - V^\pi\|_{\xi_\pi, 2}^2 \\ &\leq \gamma^2 \|\hat{V}^\pi - V^\pi\|_{\xi_\pi, 2}^2 + \|\Pi_{\phi, \xi_\pi} V^\pi - V^\pi\|_{\xi_\pi, 2}^2.\end{aligned}$$

Rearranging and taking square roots proves the bound. $\square$

However, while this result is nice, performing full Bellman update is expensive. A more practical tool is TD learning which can also be done in the approximate regime. Since we again want to focus on the distributional side, we don't mention it here and refer to [BDR23, Chapter 9.4] for more details.

2 Distributional Value Approximation

Distributional reinforcement learning replaces the scalar value function by a return-distribution function. Instead of storing only the mean $Q^\pi(x, a) = \mathbb{E}[G^\pi(x, a)]$, we keep the law of the return:

$$\eta^\pi(x, a) = \text{Law}(G^\pi(x, a)).$$

Thus

$$\eta^\pi : \mathcal{X} \times \mathcal{A} \rightarrow \mathcal{P}(\mathbb{R}),$$

where $\mathcal{P}(\mathbb{R})$ denotes probability distributions on the real line. The classical action-value function is recovered by taking expectations:

$$Q^\pi(x, a) = \int_{\mathbb{R}} z \eta^\pi(x, a)(dz).$$

The distributional Bellman operator is most cleanly written using pushforwards. For $r \in \mathbb{R}$, define the affine map

$$f_{r, \gamma}(z) = r + \gamma z.$$

If $\nu \in \mathcal{P}(\mathbb{R})$, then $(f_{r, \gamma})_\# \nu$ is the distribution of $r + \gamma Z$ when $Z \sim \nu$. The distributional Bellman operator $\mathcal{T}^\pi$ maps a return-distribution function η to

$$(\mathcal{T}^\pi \eta)(x, a) = \mathbb{E}[(f_{R, \gamma})_\# \eta(X', A') \mid X = x, A = a], \quad A' \sim \pi(\cdot \mid X').$$

The expectation averages over the random reward, the next state, and the next action sampled from the policy. The distributional Bellman equation is

$$\eta^\pi = \mathcal{T}^\pi \eta^\pi.$$

Taking means on both sides recovers the usual Bellman equation for Q^π , but the distributional equation is stronger: it describes the whole random return, not only its expectation.

Definition 8: Quantile return-distribution approximation

Just as linear TD restricts Q to a class $\mathcal{Q}_\phi$, distributional methods restrict η to a class of approximate distributions. The quantile family is one of the most important such choices. For a distribution ν with cumulative distribution function F_ν , define its quantile function by

$$F_\nu^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\nu(z) \geq \tau\}, \quad \tau \in (0, 1).$$

Choose N quantile levels, usually the midpoints

$$\tau_i = \frac{2i-1}{2N}, \quad i = 1, \dots, N.$$

An N -quantile approximation represents $\eta(x, a)$ by the empirical distribution

$$\eta_\theta(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i(x, a)},$$

where δ_z is a Dirac mass at z . The parameter $\theta_i(x, a)$ is interpreted as the predicted τ_i -quantile of the return distribution at (x, a) .

The direct distributional analogue of linear action-value approximation is to make each quantile location linear in the same feature vector:

$$\theta_i(x, a) = \phi(x, a)^\top w_i, \quad w_i \in \mathbb{R}^d.$$

Equivalently, with $W = (w_1, \dots, w_N) \in \mathbb{R}^{d \times N}$,

$$\eta_W(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\phi(x, a)^\top w_i}.$$

The scalar approximation $Q_w(x, a)$ has become a measure-valued approximation $\eta_W(x, a)$. The same features are now asked to explain not only the average return, but also the spread, asymmetry, and tail behavior of the return.

3 Quantile TD

Quantile TD applies temporal-difference learning to the quantile parameterization. For policy evaluation, sample a transition and next action

$$(x_t, a_t, r_t, x'_t, a'_t), \quad a'_t \sim \pi(\cdot \mid x'_t).$$

The distributional Bellman target is represented by the N target atoms

$$y_j = r_t + \gamma \theta_j(x'_t, a'_t), \quad j = 1, \dots, N.$$

The current prediction at (x_t, a_t) has atoms

$$z_i = \theta_i(x_t, a_t), \quad i = 1, \dots, N.$$

Definition 9: Quantile TD loss

Quantile regression uses the asymmetric absolute loss

$$\rho_\tau(u) = u(\tau - \mathbb{1}_{\{u < 0\}}).$$

The empirical Quantile TD loss is then

$$L(W) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \rho_{\tau_i}(y_j - z_i).$$

The residual $y_j - z_i$ compares an atom of the Bellman target distribution with the predicted τ_i -quantile at the current state-action pair. Minimizing this loss moves $\eta_W(x_t, a_t)$ toward

$$\frac{1}{N} \sum_{j=1}^N \delta_{r_t + \gamma \theta_j(x'_t, a'_t)} = (f_{r_t, \gamma})_{\#} \eta_W(x'_t, a'_t).$$

For linear quantiles, a subgradient update for each head is

$$w_{i,t+1} = w_{i,t} + \alpha_t \left(\frac{1}{N} \sum_{j=1}^N (\tau_i - \mathbb{1}_{\{y_j < z_i\}}) \right) \phi(x_t, a_t), \quad i = 1, \dots, N.$$

This is the exact analogue of the semi-gradient TD update:

$$\delta_t = r_t + \gamma Q_{w_t}(x'_t, a'_t) - Q_{w_t}(x_t, a_t) \rightsquigarrow \{y_j - z_i\}_{i,j=1}^N.$$

The scalar TD target is replaced by a Bellman target distribution, and the single scalar TD error is replaced by all pairwise quantile residuals.

Definition 10: Quantile Huber loss

In implementations, one often replaces ρ_τ by a quantile Huber loss. This is the smooth loss used in QR-DQN; see the treatment in [BDR23, Chapter 10]. Define

$$\mathcal{L}_\kappa(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \kappa, \\ \kappa \left(|u| - \frac{1}{2}\kappa \right), & |u| > \kappa, \end{cases}$$

and

$$\rho_\tau^\kappa(u) = \left| \tau - \mathbb{1}_{\{u < 0\}} \right| \frac{\mathcal{L}_\kappa(u)}{\kappa}.$$

The corresponding loss is

$$L_\kappa(W) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \rho_{\tau_i}^\kappa(y_j - z_i).$$

The Huber version keeps the asymmetric quantile penalty but smooths the non-differentiability around zero, which is useful when the quantile locations are produced by a neural network.

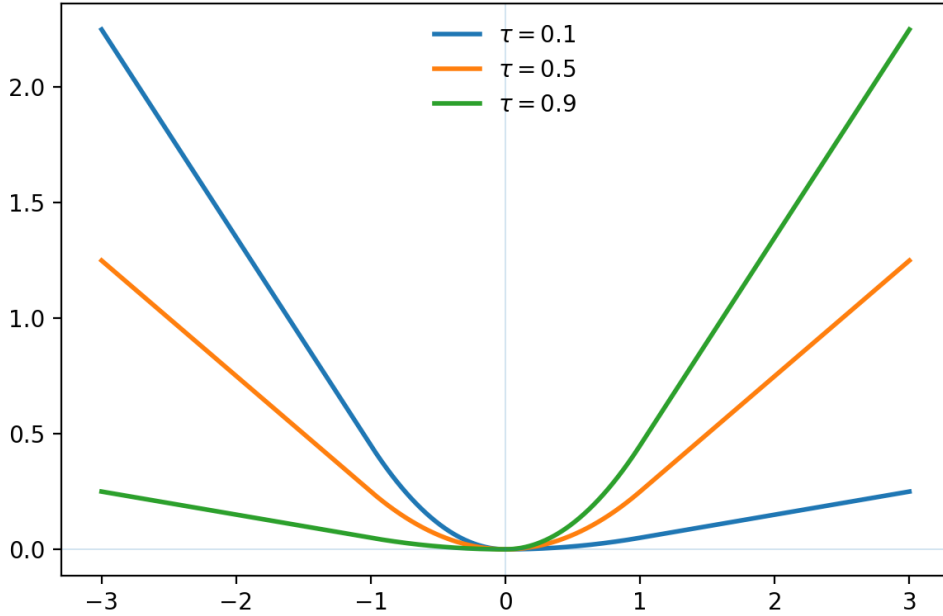


Figure 1: Quantile Huber losses for several quantile levels. The quadratic region near zero smooths the kink of the linear quantile loss while retaining its asymmetric weighting.

For control, the next action is usually selected greedily with respect to the mean of the predicted distribution:

$$a^*(x') \in \arg \max_{a' \in \mathcal{A}} \frac{1}{N} \sum_{j=1}^N \theta_j(x', a').$$

The target atoms become

$$y_j = r + \gamma \theta_j(x', a^*(x')), \quad j = 1, \dots, N.$$

This is the quantile analogue of replacing policy evaluation TD by Q-learning.

4 Gateway to Deep Representations

The linear approximation above separates representation and prediction:

$$\text{fixed features } \phi(x, a) \quad \text{and} \quad \text{linear heads } w_i.$$

This is the useful gateway into deep distributional reinforcement learning. The deep version keeps the same conceptual decomposition, but replaces the fixed feature map by a learned representation. A neural network maps the observation or state to a feature vector

$$\psi_\xi(x) \in \mathbb{R}^d,$$

and the final layers produce a distributional prediction. In the quantile case one may write

$$\theta_i(x, a; \omega) = f_i(\psi_\xi(x), a), \quad i = 1, \dots, N.$$

The return-distribution approximation becomes

$$\eta_\omega(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i(x, a; \omega)}.$$

The parameter ω includes both the representation parameters ξ and the prediction heads. Linear Quantile TD is recovered as the special case in which ψ_ξ is fixed and f_i is linear. Deep quantile methods instead optimize all of ω by backpropagating through the distributional TD loss.

5 QR-DQN

QR-DQN is the deep control version of Quantile TD. It uses a neural network to output N quantile locations for each action:

$$\theta_i(x, a; \omega), \quad i = 1, \dots, N, \quad a \in \mathcal{A}.$$

For each state-action pair, the predicted return distribution is

$$\eta_\omega(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i(x, a; \omega)}.$$

The predicted scalar action-value is the mean of this empirical distribution:

$$Q_\omega(x, a) = \frac{1}{N} \sum_{i=1}^N \theta_i(x, a; \omega).$$

Thus QR-DQN can still choose greedy actions in the usual value-based way, but the value used for action selection is obtained by averaging the learned quantiles.

As in DQN, QR-DQN uses two sets of parameters. The online network has parameters ω and is updated by gradient descent. The target network has parameters $\bar{\omega}$ and is held fixed for several updates before being copied from ω . For a transition

$$(x, a, r, x', d),$$

where $d \in \{0, 1\}$ indicates termination, the greedy next action is selected by the mean of the target-network distribution:

$$a^*(x') \in \arg \max_{a' \in \mathcal{A}} \frac{1}{N} \sum_{j=1}^N \theta_j(x', a'; \bar{\omega}).$$

The target quantile atoms are

$$y_j = r + \gamma(1 - d)\theta_j(x', a^*(x'); \bar{\omega}), \quad j = 1, \dots, N.$$

The online prediction atoms for the sampled action are

$$z_i = \theta_i(x, a; \omega), \quad i = 1, \dots, N.$$

Definition 11: QR-DQN training objective

QR-DQN minimizes the pairwise quantile Huber loss

$$L_{\text{QR}}(\omega) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \rho_{\tau_i}^{\kappa}(y_j - z_i),$$

with quantile levels

$$\tau_i = \frac{2i - 1}{2N}.$$

In a minibatch $\mathcal{B}$ sampled from replay memory, the objective is

$$\mathcal{L}_{\text{QR}}(\omega) = \frac{1}{|\mathcal{B}|} \sum_{(x, a, r, x', d) \in \mathcal{B}} L_{\text{QR}}(\omega; x, a, r, x', d).$$

The target atoms are treated as constants when differentiating with respect to ω ; this is the usual semi-gradient idea inherited from TD learning and DQN.

The quantile Huber loss is important because the raw quantile loss $\rho_{\tau}(u)$ is non-smooth at zero. QR-DQN uses

$$\mathcal{L}_{\kappa}(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \kappa, \\ \kappa \left(|u| - \frac{1}{2}\kappa \right), & |u| > \kappa, \end{cases}$$

and

$$\rho_{\tau}^{\kappa}(u) = \left| \tau - \mathbb{1}_{\{u < 0\}} \right| \frac{\mathcal{L}_{\kappa}(u)}{\kappa}.$$

For small residuals, the loss has a squared-error shape, which gives smoother gradients. For large residuals, it grows linearly, which prevents very large TD errors from dominating

the update. The factor $|\tau - \mathbb{1}_{\{u < 0\}}|$ keeps the asymmetry needed for quantile regression: underestimation and overestimation are penalized differently depending on the quantile level τ .

How QR-DQN Relates to DQN

DQN learns one scalar per action:

$$x \mapsto \{Q_\omega(x, a) : a \in \mathcal{A}\}.$$

QR-DQN learns a collection of quantiles per action:

$$x \mapsto \{\theta_i(x, a; \omega) : i = 1, \dots, N, a \in \mathcal{A}\}.$$

The scalar Bellman target

$$r + \gamma \max_{a'} Q_{\bar{\omega}}(x', a')$$

is replaced by the empirical target distribution

$$\frac{1}{N} \sum_{j=1}^N \delta_{r+\gamma(1-d)\theta_j(x', a^*(x'); \bar{\omega})}.$$

The squared TD loss is replaced by the pairwise quantile Huber loss. Action selection remains risk-neutral unless we deliberately change it: QR-DQN chooses actions by the mean of the predicted quantiles. The learned object, however, is much richer than the mean.

Representation Learning Signal

The representation $\psi_\xi(x)$ in QR-DQN is trained by many losses at once. For a single transition, the network sees all residuals

$$y_j - \theta_i(x, a; \omega), \quad i, j = 1, \dots, N.$$

Lower quantile heads receive large gradients when the lower tail is poorly placed; upper quantile heads receive large gradients when the upper tail is poorly placed. Thus the shared representation is shaped not only by expected return, but by the whole geometry of the target return distribution.

This gives a useful interpretation of the empirical strength of distributional methods. A scalar DQN representation only needs to make the conditional mean easy to predict. A QR-DQN representation must make many parts of the return distribution easy to predict simultaneously. The extra prediction problem can act as a richer training signal, even when the final policy is still selected by the mean.

6 Particle Distributional DQN

QR-DQN uses particles with a special interpretation: the particle $\theta_i(x, a)$ is tied to the fixed quantile level $\tau_i = (2i - 1)/(2N)$. MMD-DQN and SinkhornDRL keep the empirical-measure

form

$$\eta_\omega(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{z_i(x, a; \omega)}$$

but treat the z_i as deterministic samples of the return distribution, not as named quantiles. The common template is:

1. choose a divergence or metric d between probability measures,
2. study whether the distributional Bellman operator contracts under the induced supremum metric

$$\bar{d}(\eta, \nu) = \sup_{x, a} d(\eta(x, a), \nu(x, a)),$$

3. use the empirical version of d as a differentiable DQN loss.

The last point matters algorithmically. The ordinary Wasserstein distance is mathematically natural for distributional RL, but its empirical optimization with unrestricted particles is not as directly smooth as the losses below.

6.1 MMD-DQN

Definition 12: Kernel mean embedding and maximum mean discrepancy

Let $k : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous positive-definite kernel and let $\mathcal{H}_k$ be its reproducing kernel Hilbert space. For a probability measure μ , define the kernel mean embedding

$$\psi_\mu = \int k(z, \cdot) \mu(dz) \in \mathcal{H}_k.$$

The maximum mean discrepancy is

$$\text{MMD}_k(\mu, \nu) = \|\psi_\mu - \psi_\nu\|_{\mathcal{H}_k}.$$

Equivalently, if $Z, Z' \sim \mu$ and $Y, Y' \sim \nu$ are independent, then

$$\text{MMD}_k^2(\mu, \nu) = \mathbb{E}k(Z, Z') + \mathbb{E}k(Y, Y') - 2\mathbb{E}k(Z, Y).$$

If the kernel is characteristic, the embedding is injective, and $\text{MMD}_k(\mu, \nu) = 0$ implies $\mu = \nu$. Thus the MMD is then a genuine metric on probability measures.

Theorem 4: When MMD is a metric [NTGV21, Theorem 1]

A characteristic kernel, such as the Gaussian kernel, induces an MMD metric on $\mathcal{P}(\mathbb{R})$. The unrectified distance kernel

$$k_\alpha(x, y) = -|x - y|^\alpha$$

induces an MMD metric on distributions with finite α -moment for $0 < \alpha < 2$, but not for $\alpha = 2$. The exponential-product kernel $k(x, y) = \exp(xy/\sigma^2)$ is likewise a metric on the paper's moment-determinate domain $\mathcal{P}_*(\mathbb{R})$.

Nguyen et al. define the state-action supremum metric [NTGV21]

$$\text{MMD}_{\infty,k}(\eta, \nu) = \sup_{x,a} \text{MMD}_k(\eta(x, a), \nu(x, a)).$$

The contraction theorem does not hold for arbitrary characteristic kernels. The clean theorem assumes kernels that are translation invariant and homogeneous under discount scaling in the following sense:

$$k(x + c, y + c) = k(x, y)$$

and, for $b_{\gamma,0}(z) = \gamma z$,

$$\text{MMD}_k((b_{\gamma,0})_{\#}\mu, (b_{\gamma,0})_{\#}\nu) = \gamma^{\alpha/2} \text{MMD}_k(\mu, \nu).$$

Theorem 5: MMD Bellman contraction and its limitation [NTGV21, Theorem 2]

For $k = \sum_{i \in I} c_i k_i$, where $c_i \geq 0$ and every k_i is translation invariant and homogeneous of order $\alpha_i > 0$, set $\alpha_* := \inf_{i \in I} \alpha_i$. The distributional Bellman operator satisfies

$$\text{MMD}_{\infty,k}(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \nu) \leq \gamma^{\alpha_*/2} \text{MMD}_{\infty,k}(\eta, \nu).$$

Hence the metric-first story mirrors the Wasserstein contraction story: if the chosen metric contracts under the affine Bellman shift and mixing does not expand it, fixed-point arguments become available. The operator is not, however, a contraction for Gaussian or exponential-product kernels in general. Thus characteristicness alone is not enough for Bellman contraction.

Theorem 6: Deterministic-particle approximation rate [NTGV21, Theorem 3]

Let $P \in \mathcal{P}(\mathbb{R}^d)$ and choose n deterministic particles to minimize MMD,

$$P_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}, \quad (x_i)_{i=1}^n \in \arg \min_{(x_i)_{i=1}^n} \text{MMD}_k \left(\frac{1}{n} \sum_{i=1}^n \delta_{x_i}, P \right).$$

For every h in the unit ball of the corresponding RKHS,

$$\left| \int h dP_n - \int h dP \right| = O(n^{-1/2}).$$

This rate is independent of the ambient dimension; the paper contrasts it with the usual $O(n^{-1/d})$ empirical 1-Wasserstein rate when $d > 2$.

The important caveat is exactly the one emphasized in the presentation. The kernel used for the clean theorem is not necessarily the kernel that performs best empirically. Kernels such as

$$k_\alpha(x, y) = -\|x - y\|^\alpha$$

fit the contraction theory, whereas Gaussian kernels generally do not yield a Bellman con-

traction in MMD. Nevertheless, Gaussian kernels can perform better in experiments, so the theorem should be read as a stability result for a class of metrics, not as a full explanation of the best empirical kernel choice.

For a transition (x, a, r, x') and target-network parameters $\bar{\omega}$, the particle DQN target is constructed by first choosing the greedy next action

$$a^*(x') \in \arg \max_{a' \in \mathcal{A}} \frac{1}{N} \sum_{j=1}^N z_j(x', a'; \bar{\omega})$$

and then shifting the target particles:

$$y_j = r + \gamma z_j(x', a^*(x'); \bar{\omega}), \quad j = 1, \dots, N.$$

With current particles $z_i = z_i(x, a; \omega)$, the empirical MMD loss is

$$L_{\text{MMD}}(\omega) = \frac{1}{N^2} \sum_{i, i'=1}^N k(z_i, z_{i'}) + \frac{1}{N^2} \sum_{j, j'=1}^N k(y_j, y_{j'}) - \frac{2}{N^2} \sum_{i, j=1}^N k(z_i, y_j).$$

Only the terms involving current particles are differentiated with respect to ω ; the target particles are fixed by the target network.

6.2 SinkhornDRL

Definition 13: Regularized optimal transport and Sinkhorn divergence

SinkhornDRL starts from optimal transport and replaces the hard transport problem by an entropically regularized one [Sun+24]. For a cost $c : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_+$, define

$$\text{OT}_c(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c(z, y) \pi(\text{d}z, \text{d}y),$$

where $\Pi(\mu, \nu)$ is the set of couplings with marginals μ and ν . The entropic version is

$$\text{OT}_c^\varepsilon(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \left\{ \int c(z, y) \pi(\text{d}z, \text{d}y) + \varepsilon \text{KL}(\pi \mid \mu \otimes \nu) \right\}.$$

Because this regularized objective has a self-similarity bias, SinkhornDRL uses the de-biased Sinkhorn divergence

$$\bar{W}_{c, \varepsilon}(\mu, \nu) = \text{OT}_c^\varepsilon(\mu, \nu) - \frac{1}{2} \text{OT}_c^\varepsilon(\mu, \mu) - \frac{1}{2} \text{OT}_c^\varepsilon(\nu, \nu).$$

The self-cost correction removes the regularized objective's self-similarity bias while retaining a smooth, computationally tractable loss.

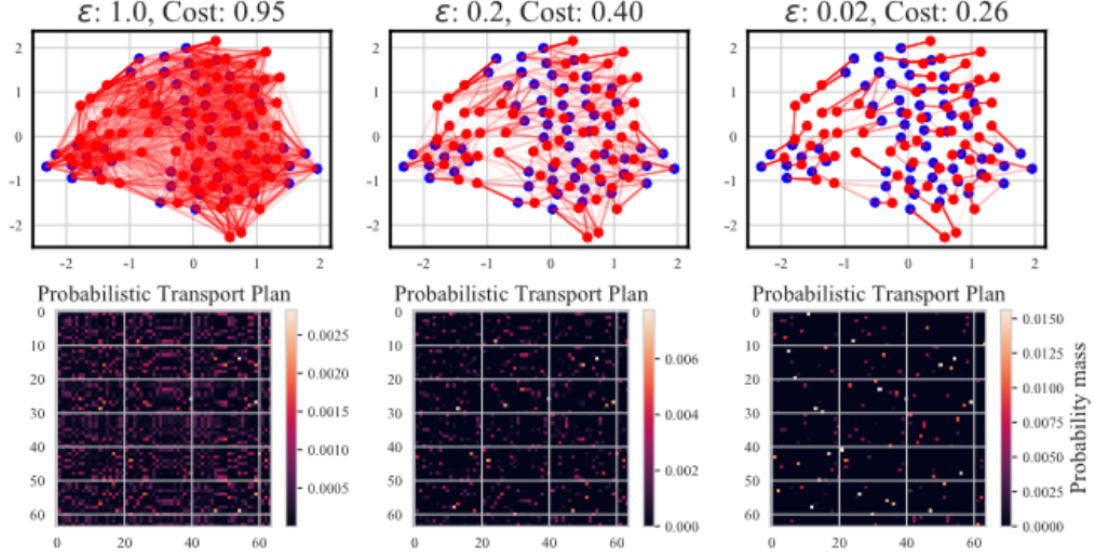


Figure 2: Entropic regularization controls the diffuseness of the transport plan: decreasing ε concentrates mass on a sparser matching.

Proposition 1: Non-expansion of the information term [Sun+24, Proposition 1]

For a nontrivial coupling Π , define

$$\text{MI}_{\Pi}^{\infty}(\eta, \nu) = \sup_{x,a} \text{KL}(\Pi_{x,a} \mid \eta(x, a) \otimes \nu(x, a)).$$

Then the distributional Bellman operator is nonexpansive:

$$\text{MI}_{\Pi}^{\infty}(\mathcal{T}^{\pi}\eta, \mathcal{T}^{\pi}\nu) \leq \text{MI}_{\Pi}^{\infty}(\eta, \nu).$$

The regularization component therefore does not expand under a Bellman update, but it does not supply the strict discount contraction of the transport cost.

For $c(x, y) = \|x - y\|^{\alpha}$, define the supremum divergence

$$\overline{W}_{c,\varepsilon}^{\infty}(\eta, \nu) = \sup_{x,a} \overline{W}_{c,\varepsilon}(\eta(x, a), \nu(x, a)).$$

Theorem 7: Sinkhorn Bellman contraction [Sun+24, Theorem 1]

Sun et al. show that the Bellman operator is contractive with a factor bounded by

$$\overline{\Delta}_\varepsilon(\gamma, \alpha) = \gamma^\alpha \left(1 - \sup_{\mu, \nu} \lambda_\varepsilon(\mu, \nu) \right) + \sup_{\mu, \nu} \lambda_\varepsilon(\mu, \nu),$$

where

$$\lambda_\varepsilon(\mu, \nu) = \frac{\varepsilon \text{KL}(\pi^* \mid \mu \otimes \nu)}{\overline{W}_{c, \varepsilon}(\mu, \nu)} \in (0, 1)$$

and π^* is an optimal regularized coupling. Thus

$$\overline{W}_{c, \varepsilon}^\infty(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \nu) \leq \overline{\Delta}_\varepsilon(\gamma, \alpha) \overline{W}_{c, \varepsilon}^\infty(\eta, \nu).$$

The rate lies between the unregularized Wasserstein-type rate and the regularization-dominated rate:

$$\overline{\Delta}_\varepsilon(\gamma, \alpha) \in (\gamma^\alpha, 1).$$

The particle learning rule has the same Bellman target construction as MMD-DQN. With current empirical measure

$$\eta_\omega(x, a) = \frac{1}{N} \sum_{i=1}^N \delta_{z_i(x, a; \omega)}$$

and target empirical measure

$$\hat{\eta}_{\bar{\omega}}(x, a, r, x') = \frac{1}{N} \sum_{j=1}^N \delta_{r + \gamma z_j(x', a^*(x'); \bar{\omega})},$$

the loss is

$$L_{\text{RW}}(\omega) = \overline{W}_{c, \varepsilon}(\eta_\omega(x, a), \hat{\eta}_{\bar{\omega}}(x, a, r, x')).$$

The Sinkhorn algorithm gives a differentiable approximation to this quantity for empirical measures, which makes it usable as a neural-network loss.

Theorem 8: Wasserstein–MMD interpolation [Sun+24, Theorem 1]

The regularization parameter interpolates between two geometries:

$$\varepsilon \rightarrow 0 \quad \Rightarrow \quad \overline{W}_{c, \varepsilon}(\mu, \nu) \rightarrow 2W_c^\alpha(\mu, \nu),$$

whereas

$$\varepsilon \rightarrow +\infty \quad \Rightarrow \quad \overline{W}_{c, \varepsilon}(\mu, \nu) \rightarrow \text{MMD}_{k_\alpha}^2(\mu, \nu), \quad k_\alpha = -c.$$

Thus the small-regularization endpoint recovers transport geometry, while the large-regularization endpoint recovers the MMD induced by the negative cost kernel.

This is why the presentation treats SinkhornDRL as a bridge between Wasserstein-style distributional RL and MMD-DQN: it keeps a transport-based objective while making the empirical particle loss smooth enough for deep learning.

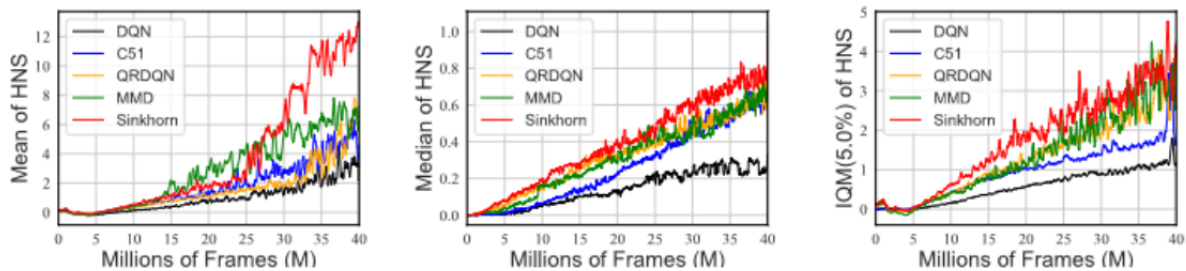


Figure 3: Atari aggregate performance reported in the presentation. The three panels show the mean, median, and interquartile mean of human-normalized scores over training.

References

- [BDR23] Marc G. Bellemare, Will Dabney, and Mark Rowland. *Distributional Reinforcement Learning*. MIT Press, 2023.
- [NTGV21] Thanh Nguyen-Tang, Sunil Gupta, and Svetha Venkatesh. “Distributional Reinforcement Learning via Moment Matching”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 35. 10. 2021, pp. 9144–9152. DOI: 10.1609/aaai.v35i10.17104.
- [Sun+24] Ke Sun et al. “Distributional Reinforcement Learning with Regularized Wasserstein Loss”. In: *Advances in Neural Information Processing Systems*. Vol. 37. 2024.